Functions

ANSWER KEY

Function notation and evaluation

1 If $f(x) = 2x^2 - 3x + 1$, find $f(4)$.

$f(4) = 2(16) - 3(4) + 1 = 32 - 12 + 1 = 21$.

2 If $g(x) = \frac{x+3}{x-1}$, find $g(-2)$.

$g(-2) = \frac{-2+3}{-2-1} = \frac{1}{-3} = -\frac{1}{3}$.

Domain and range

3 Find the domain of $f(x) = \frac{1}{x-5}$.

The function is undefined when the denominator is zero: $x - 5 = 0$, so $x = 5$. Domain: all real numbers except $x = 5$.

4 Find the domain of $g(x) = \sqrt{x-3}$.

The expression under the square root must be non-negative: $x - 3 \geq 0$, so $x \geq 3$. Domain: $x \geq 3$.

Composite functions

5 If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $f(g(2))$.

First find $g(2) = 2(2) - 3 = 1$. Then $f(1) = 1^2 + 1 = 2$.

6 If $f(x) = 3x - 2$ and $g(x) = x + 4$, find $(g \circ f)(x)$ as a simplified expression.

$(g \circ f)(x) = g(f(x)) = g(3x - 2) = (3x - 2) + 4 = 3x + 2$.

Inverse functions

7 Find the inverse of $f(x) = 3x - 6$.

Let $y = 3x - 6$. Swap x and y: $x = 3y - 6$. Solve for y: $x + 6 = 3y$, so $y = \frac{x+6}{3}$. Thus $f^{-1}(x) = \frac{x+6}{3}$.

8 If $f(5) = 12$, find $f^{-1}(12)$.

By the definition of an inverse function, $f^{-1}(12) = 5$, since inverse functions reverse input and output pairs.

Transformations of functions

9 The graph of $y = f(x)$ is shifted 3 units right and 2 units up. Write the equation of the transformed function in terms of f.

Shifting right by 3 replaces x with $x - 3$; shifting up by 2 adds 2 to the output. Transformed function: $y = f(x - 3) + 2$.

10 Describe the transformation that takes $y = x^2$ to $y = -2(x + 1)^2 + 5$.

The graph is reflected over the x-axis and vertically stretched by a factor of 2 (from the -2), shifted 1 unit left (from $x + 1$), and shifted 5 units up (from $+5$).

Piecewise functions

11 A function is defined as $f(x) = x + 2$ for $x < 3$ and $f(x) = 2x - 1$ for $x \geq 3$. Find $f(1)$ and $f(5)$.

Since $1 < 3$, use the first piece: $f(1) = 1 + 2 = 3$. Since $5 \geq 3$, use the second piece: $f(5) = 2(5) - 1 = 9$.

Interpreting function graphs and increasing/decreasing behavior

12 A function's graph rises from $x = 0$ to $x = 3$, reaches a peak at $x = 3$, then falls from $x = 3$ to $x = 6$. State the intervals where the function is increasing and decreasing, and identify the type of point at $x = 3$.

The function is increasing on the interval $0 < x < 3$ and decreasing on the interval $3 < x < 6$. The point at $x = 3$ is a local maximum, since the function's value is greater there than at nearby points on both sides.

Modeling with function families

13 A company's profit is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the number of units sold (in hundreds). Determine whether this is a linear, quadratic, or exponential model, and find the number of units that maximizes profit.

This is a quadratic model, identifiable by the x^2 term. The maximum occurs at $x = -\frac{b}{2a} = -\frac{40}{2(-2)} = 10$, meaning profit is maximized when 1000 units (10 hundreds) are sold.

Finding function values from a graph or table

14 A table shows: $f(0) = 3$, $f(1) = 5$, $f(2) = 9$, $f(3) = 15$. Determine whether this could represent a linear or quadratic function, by examining the differences between consecutive values.

First differences: $5 - 3 = 2$, $9 - 5 = 4$, $15 - 9 = 6$. Since the first differences are not constant (they increase by 2 each time), the function is not linear. Second differences: $4 - 2 = 2$, $6 - 4 = 2$ — these ARE constant, which is the signature of a quadratic function.

Zeros of a function and their graphical meaning

15 A function $f(x) = (x - 2)(x + 5)$ has zeros where it crosses the x-axis. Find the zeros, and state the y-intercept of the function.

Zeros occur where each factor equals zero: $x = 2$ and $x = -5$. The y-intercept is found at $x = 0$: $f(0) = (0 - 2)(0 + 5) = (-2)(5) = -10$, so the y-intercept is $(0, -10)$.

Even and odd functions

16 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, by evaluating $f(-x)$ and comparing to $f(x)$.

$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$. Since $f(-x) = f(x)$, the function is even (its graph is symmetric about the y-axis).

17 Determine whether $g(x) = x^3 - 2x$ is even, odd, or neither.

$g(-x) = (-x)^3 - 2(-x) = -x^3 + 2x = -(x^3 - 2x) = -g(x)$. Since $g(-x) = -g(x)$, the function is odd (its graph has point symmetry about the origin).

Modeling a real-world quantity with a function and evaluating it

18 The cost to rent a car is modeled by $C(d) = 35 + 0.20d$, where d is the number of miles driven. Find $C(150)$ and interpret the result in context.

$C(150) = 35 + 0.20(150) = 35 + 30 = 65$. This means renting the car and driving 150 miles costs 65 dollars total.

Solving for an input given an output

19 If $f(x) = 2x - 7$ and $f(a) = 15$, find the value of a.

$2a - 7 = 15$, so $2a = 22$, giving $a = 11$.

20 If $g(x) = x^2 - 4$ and $g(b) = 21$, find all possible values of b.

$b^2 - 4 = 21$, so $b^2 = 25$, giving $b = 5$ or $b = -5$.